

Contact

9822034387 (Mobile)
akkeshavan@gmail.com

www.linkedin.com/in/anand-kumar-keshavan-18829a1 (LinkedIn)

Top Skills

Concept Generation
Emerging Technologies
Autogen/Langgraph Multi-agent systems

Publications

The Magical World Of Functional Programming -Part I
Deep Dive. The Magical World of Functional Programming Part II

Patents

Wireless Payment System And Multi-Factor Authentication(MFA)

Anand Kumar Keshavan

Software Developer| Researcher | Computational Linguistics
| Generative AI| Formal Methods | Author | Musician(Retired)|
Neurodivergent
Pune, Maharashtra, India

Summary

After a long and successful career building software related businesses, where I held top management positions in several companies, I have returned to my original passion- creating software that is rooted in the fundamental principles of computer science.

Consultant for architecting new products , and re-engineering existing ones for massive scalability, maintainability and fault tolerance, using advanced concurrent/functional programming techniques.

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Highlights:

- Author of two books on Functional Programming
- Re-engineering and migration of one of the most popular charting libraries to Javascript
- Creating the system integration team for one of the earliest Linux based Operating systems for mobile phones (~2007)
- Creating the team and re-architecting the technology stacks for one of the world's largest poker networks (~2006)
- Leading the creation of semiconductor industry practice for a leading services company.(~2001-5)
- Invented the multi-factor authentication and related aspects of wireless payment systems.(2000)
- Architected one of the earliest Object oriented languages/ development toolchain that was targeted at business applications . (1992-1995)
- One of the first SLR compiler-compiler systems for MS-DOS (1980s)
- One of the earliest text search engines and NLP system (1980s)

Worked across various countries such as US, UK, Europe and Japan.Possess domain knowledge in System software (compilers,

operating system, embedded software, mobile, large scale applications). Rich experience in creating great product development teams.

Publications:

1. The Magical World Of Functional Programming -Part I: Thinking Functional
2. Deep Dive. The Magical World Of Functional Programming- Part II

GitHub: <https://github.com/akkeshavan>

Experimental Projects:

Spectre: A formal specification language.

<https://github.com/BoundlessProducts/spectre>

Hand on polyglot programmer.

Languages : Node.JS, Javascript/TypeScript, Python

Research languages: Rust, Haskell

Formal Specification, Formal

Verification using TLA+

GenAI: RAG pipelines with LLMs, Fine-tuning Huggingface and other models. Multi Agent (Agentic) systems using Autogen, LangGraph, own custom framework.

Experience

Postman

Principal Engineer at Postman labs

July 2022 - Present (3 years 8 months)

Pune, Maharashtra, India

Solving hard problems using computer science in development of the "zero-to-one" products in Postman Labs. Involved in research, architecture, development and coaching.

Self-employed

Research Projects (On-going)

January 2025 - Present (1 year 2 months)

Pune, Maharashtra, India

As a lifelong researcher, I am curious about many subjects and go where my curiosity takes me!!

Here are the links to some of my projects:

1. Building a compiler using Rust and LLVM

<https://insatiable-curiosity.gitbook.io/building-a-compiler-with-llvm-and-rust-a-tutorial>

2. The First programming language - a functional typed functional language

<https://github.com/akkeshavan/first-book>

3. Spectre - a Formal specification and verification tool

<https://github.com/BoundlessProducts/spectre>

Extentia Information Technology

Chief Technical Architect

June 2019 - May 2022 (3 years)

Pune Area, India

Charting an ambitious course for meeting the Engineering Excellence targets of Extentia.

The role involves building a strong culture of learning and inculcating a rigor amongst the engineering teams of Extentia.

Architecting complex systems that can be extended, scaled, and maintained in the long term.

Conducting rigorous reviews of all technical artifacts- code, design and architecture, by creating reference implementations.

SwanSpeed Consulting

Founder/Partner

January 2018 - June 2019 (1 year 6 months)

Pune

Self Employed

5 years 1 month

Independent consultant for product development companies, Director Codewalla

December 2013 - January 2018 (4 years 2 months)

Co-Founder

January 2013 - January 2018 (5 years 1 month)

Pune Area, India

Energy management , smart meters solutions

Searce

Vice-President(Product Development)

October 2012 - November 2013 (1 year 2 months)

Working on next generation of SAAS products. The first product to be rolled out will be an HR management software to be offered as a service on the Google App engine platform. The product has a contemporary UX and rich functionality while being easy to set-up and roll-out.

FusionCharts

VP-Engineering

August 2011 - September 2012 (1 year 2 months)

Bangalore

Successfully shipped FusionCharts XT Suite of products which consists of FusionCharts XT, FusionWidgets XT, PowerCharts XT and FusionMaps XT. The program resulted in the conversion of the existing Flash based product suite to JavaScript. Also shipped the first commercial version of Collabion Charts for Sharepoint.

Please visit www.fusioncharts.com for details of products shipped as a part of this gig!

Thoughtworks India Private Ltd

General Manager

April 2010 - September 2010 (6 months)

Gurgaon

Azingo (Now a part of Motorola Mobility, A Google Co.)

Senior Director-Services

August 2007 - April 2009 (1 year 9 months)

Pune

Responsible for setting up the teams that created the earliest commercial releases of Azingo's linux based mobile operating system and components. Customers included Tier-1 handset manufacturers and chipmakers.

Company eventually taken over by Motorola Mobility, now a part of Google Inc

Indecca technologies

VP - Engg

2006 - 2007 (1 year)

Pune

Set up the engineering organization for one of the largest online gaming technology companies specializing in online poker.

Kpit Cummins Infosystems Ltd

General Manager- Advanced Technology Solutions(Semiconductors)
July 2001 - July 2005 (4 years 1 month)

Responsible for setting up the Semiconductr Industry Solutions consisting of providing services in VLSI, Embedded Software, Compilers and Tools and applications.

Also set up the Japanese operations for the company.

Ruksun Software Technologies

Engineering Manager-> VP-> President
April 1995 - July 2001 (6 years 4 months)

After joining in 1995 as Engineering, was responsible for setting the initial enigneering organziation for the company. follwed by several roles that included managing all various functions of a software organization. Served as President of the company responsible for all operations for a year.

Represented the company's interests in two subsidiaries/associates by serving on the board of Directors of Ruksun Doctoranywhere.com and Ruksun EasyDiary.com.

Doctoranywhere.com

Member on the Board of Directors
1999 - 2001 (2 years)

Pune

Nominee director for Ruksun Software Technologies (p) Ltd on the board of Ruksun DoctorAnywhere.com (p) limited, a telemedicine company that was promoted by the former.

Particpated in strategy development and fund raising initiatives.

Mastek Ltd

Manager (R&D)
1992 - 1995 (3 years)

Robsonia

Programmer

1983 - 1991 (8 years)

Education

University of Delhi

B. Com (Hons), Business · (1979 - 1982)